



Classification:



Analyzing Sentiment

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Predicting sentiment by topic:

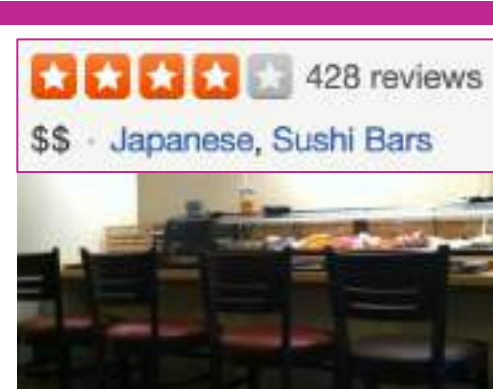
An intelligent restaurant review system

It's a big day & I want to book a table at a nice Japanese restaurant

Seattle has many
★★★★
sushi restaurants

What are people
saying about
the food?
the ambiance?...





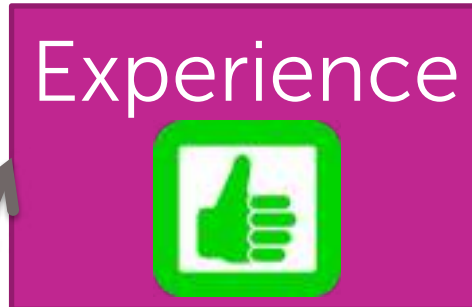
Positive reviews not positive about everything

Sample review:

Watching the chefs create incredible edible art made the experience very unique.

My wife tried their ramen and it was pretty forgettable.

All the sushi was delicious!
Easily best sushi in Seattle.



From reviews to topic sentiments

All reviews
for restaurant

★★★★☆ 1/2/2016
This is probably my favorite place to eat Japanese in Seattle. My boyfriend and I ordered night of scallops, Japanese miso soup (sashimi), and the appetizer: toki and 2 special rolls. I would skip the special rolls because the night and sashimi rolls are where the place excels. The toki, as recommended by other reviewers was amazing. It's more chewy and the sashimi is to the perfect amount of flavor for the delicate toki.

★★★★★ 6/1/2015
Dining here at the sushi bar made me feel like sitting front row to an amazing performance. We didn't have much, banged down to the 10 After work, got here traditionally at 5:15pm, and got the last two seats in the place.

★★★★★ 6/6/2016
I came here having high expectations due to the reviews of this place, but I was let disappointed. The restaurant is small so do make reservations when you come here. Dishes cost from \$5-16 each and dishes are small.

Novel intelligent
restaurant review app

Experience
★★★★

Ramen
★★★

Sushi
★★★★★

Easily best sushi
in Seattle.

Intelligent restaurant review system

All reviews for restaurant

★★★★☆ 0/5 100113
This is probably my favorite place to eat Japanese in Seattle. My boyfriend and I ordered night of scallops, Japanese miso soup (seasonal), and the seasonal fish and 2 special rolls. I would skip the special rolls because the night and season rolls is where the plate excels. The food, as recommended by other reviewers was amazing. It's more of a mix and the authenticity is the perfect amount of flavor for the delicate fish.

★★★★★ 0/5 100113
Dining here at the sushi bar made me feel like sitting front row to an amazing performance. We didn't have much, banged down to the 10 After work, got here traditionally at 8:15pm, and got the last two seats in the place.

★★★★★ 0/5 100113
I came here having high expectations due to the reviews of this place, but I was let disappointed. The restaurant is small so do make reservations when you come here. Dishes cost from \$5-10 each and dishes are small.

Break all reviews into sentences

The seaweed salad was just OK,
vegetable salad was just ordinary.

I like the interior decoration and
the blackboard menu on the wall.

All the sushi was delicious.

My wife tried their ramen and
it was pretty forgettable.

The sushi was amazing, and
the rice is just outstanding.

The service is somewhat hectic.

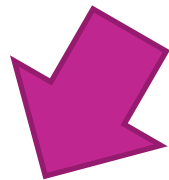
Easily best sushi in Seattle.

Core building block

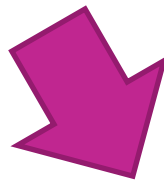
Easily best sushi in Seattle.



Sentence Sentiment
Classifier



Easily best sushi in Seattle.



Intelligent restaurant review system

All reviews
for restaurant

★★★★☆
This is probably my favorite place to eat Japanese in Seattle. My boyfriend and I ordered night of scallops, Japanese omelette (sashimi), and the appetizer: sobu and 2 special rolls. I would skip the special rolls because the night and custom rolls is where the place excels. The food, as recommended by other reviewers was amazing. It's more of a fancy and the authenticity is the perfect amount of fusion for the authentic roll.

★★★★★
All the sushi was delicious. Dining here at the sushi bar made me feel like sitting front row to an amazing performance. We didn't have much, banged down to the 10 After work, got here traditionally at 6:15pm, and got the last two seats in the place.

★★★★★
The sushi was amazing, and the rice is just outstanding. The service is somewhat hectic.

Easily best sushi in Seattle.

Break all reviews
into sub-
sentences

The seaweed salad was just OK, vegetable salad was just ordinary.

Like the interior decoration and blackboard menu on the wall.

All the sushi was delicious.

My wife tried their ramen and it was pretty forgettable.

The sushi was amazing, and the rice is just outstanding.

The service is somewhat hectic.

Easily best sushi in Seattle.

Sentence
Sentiment
Classifier

Average
predictions

Sushi
★★★★★

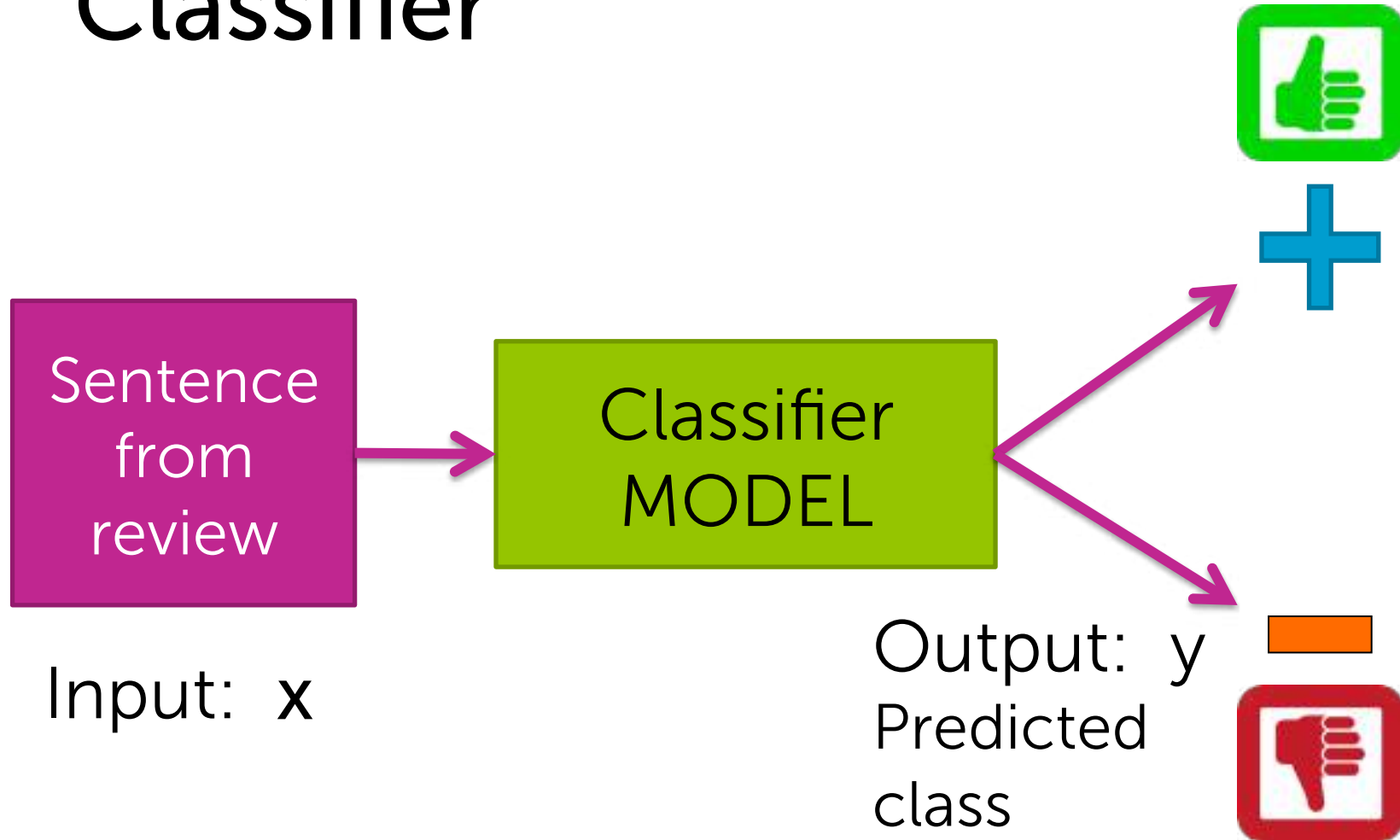
Most



Easily best
sushi
in Seattle.

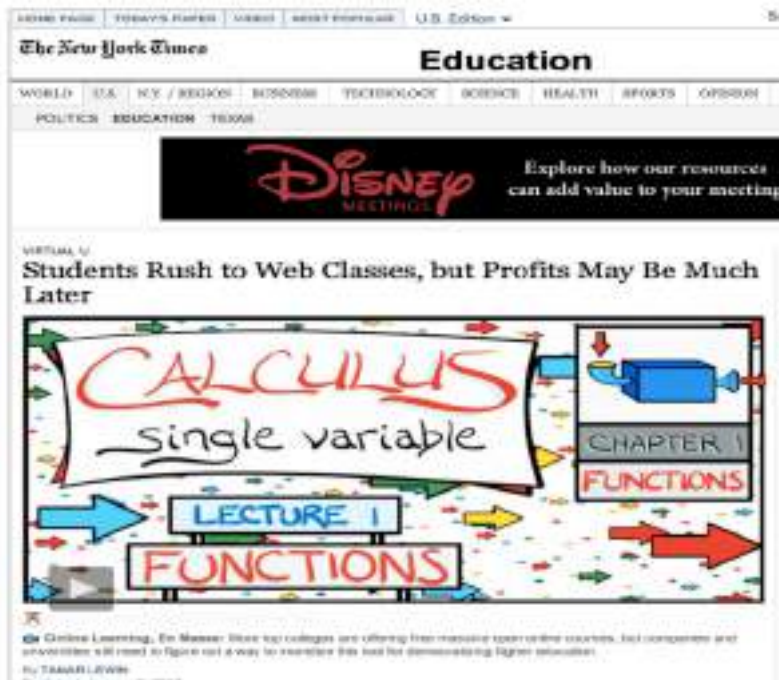
Classifier applications

Classifier



Example multiclass classifier

Output y has more than 2 categories



Education

Finance

Technology

Input: x
Webpage

Output: y

Spam filtering

German Khan to Carlos
Hi Carlos,
Let's try to chat on Friday or Saturday to coordinate our room on Tuesday's presentation.
Carlos

Welcome to New Media Installation: Art that Learns
Hi everyone,
Welcome to New Media Installation: Art that Learns.
This class will start tomorrow.
Make sure you attend the first class, even if you are on the West Coast!
The classes are held in Gallery 1001-1010, and will be Tue, Thu 5:15-6:30 PM.
By now, you should be subscribed to our course mailing list: NMIS15@stanford.edu.
You can check the instructions by emailing: NMIS15@stanford.edu.

Natural LoseWeight SuperFood Endorsed by Oprah Winfrey. Free Trial 1 bottle, pay only \$5.95 for shipping mfer nk
Dear Sir/Madam,
Natural Weight Loss Solution
Vital Vital is a natural Weight Loss product that enables people to lose weight and cleanse their bodies faster than most other products on the market.
Here are some of the benefits of Vital Vital that you might not be aware of. These benefits have helped people who have been using Vital Vital daily to achieve goals and reach new heights in their dating that they never thought they could.
* Rapid Weight Loss
* Increased Metabolism - that fat is coming easily!
* Better Mood and Attitude
* More Sex Satisfaction
* Cleanse and Detoxify Your Body
* Boost Your Energy

Not spam

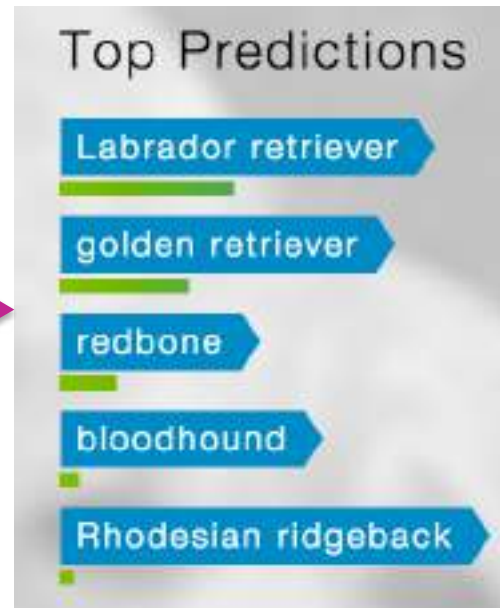
Spam

Input: x

Output: y

Text of email,
sender, IP,...

Image classification



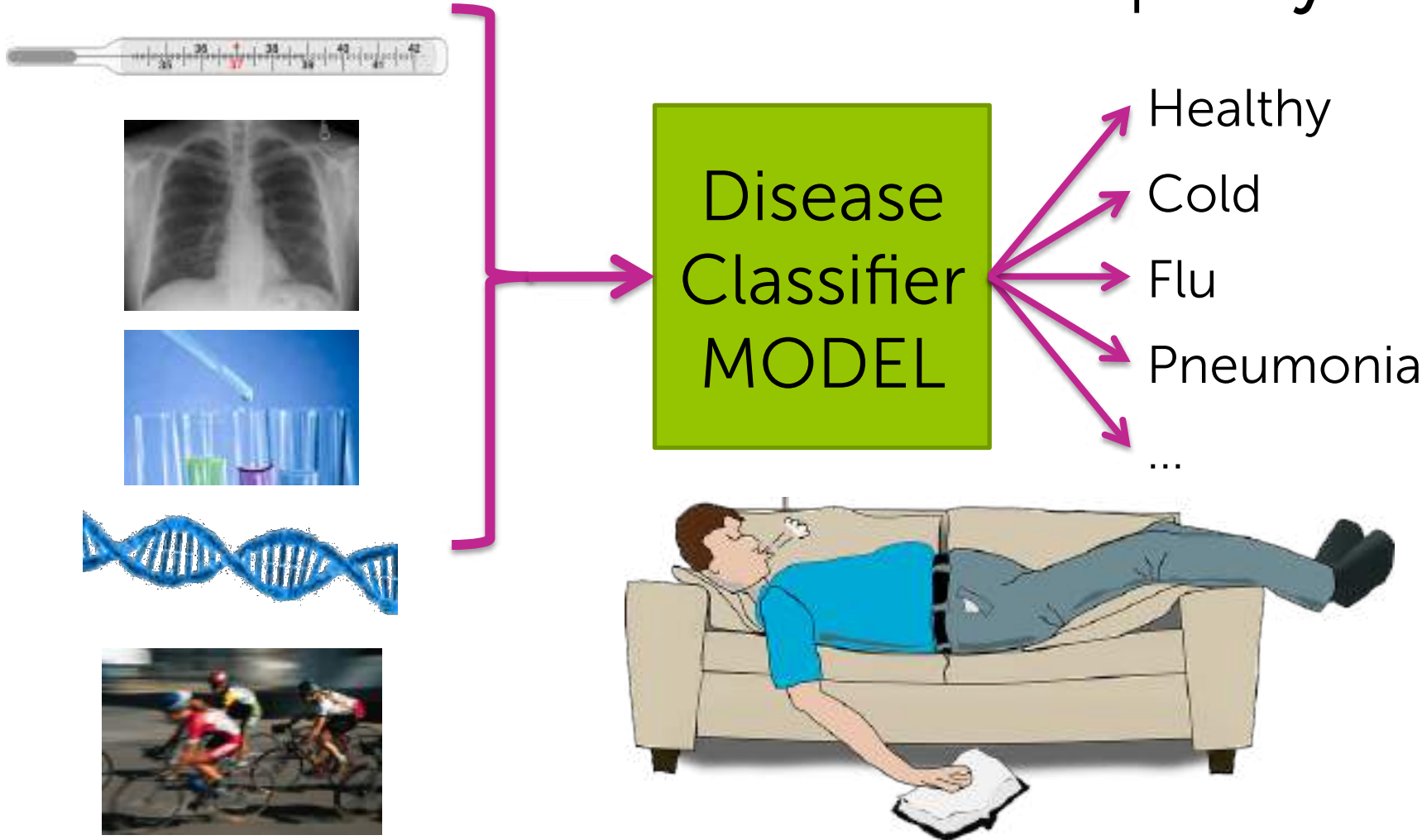
Input: x
Image pixels

Output: y
Predicted object

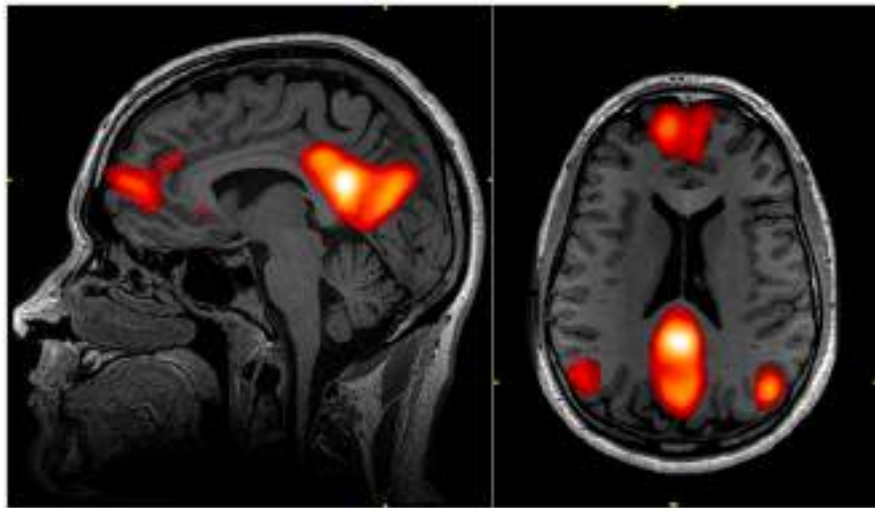
Personalized medical diagnosis

Input: x

Output: y



Reading your mind



"Hammer"

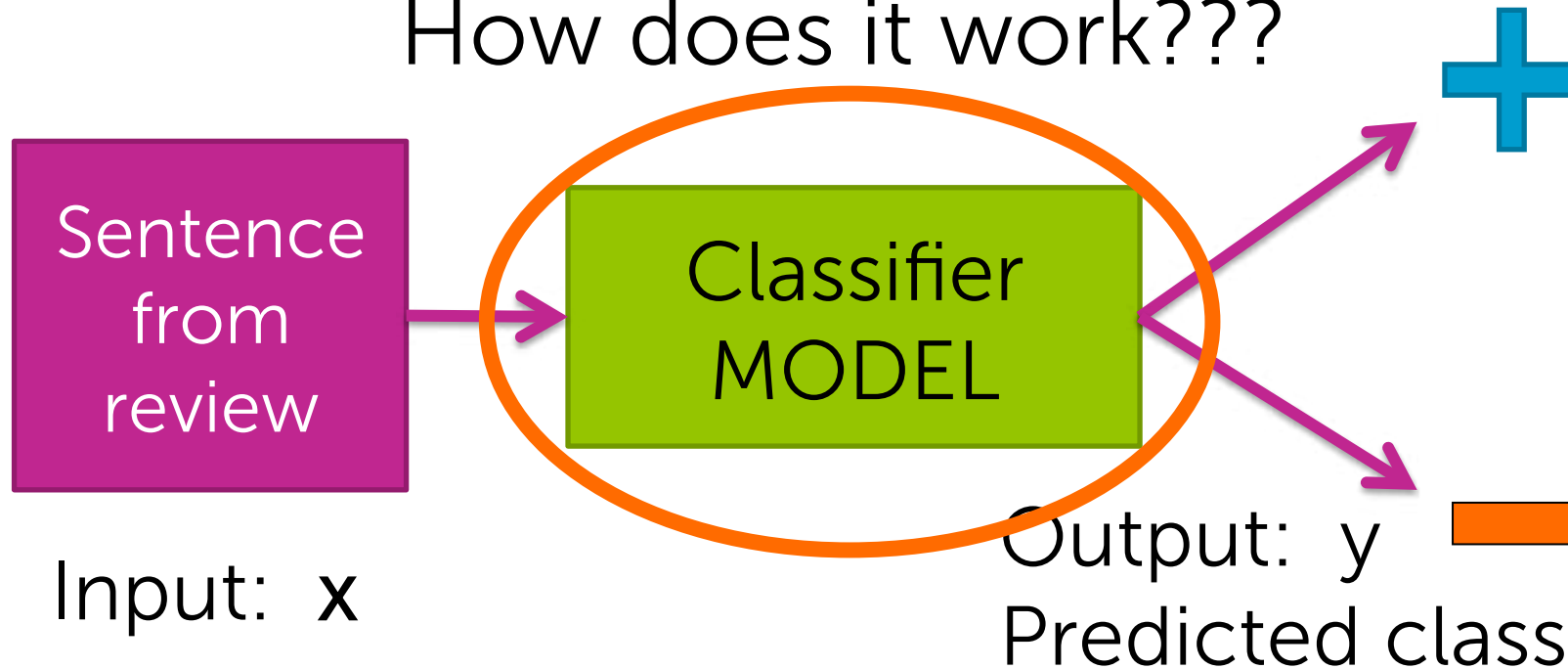


"House"

Linear classifiers

Representing classifiers

How does it work???



List of positive
words

great, awesome,
good, amazing,...

List of negative
words

bad, terrible,
disgusting, sucks,...



Simple threshold classifier

Count positive & negative words
in sentence

Sentence
from
review



If *number of positive words* >
number of negative words:

$\hat{y} =$



Else:

$\hat{y} =$



Input: x

List of positive words

great, awesome,
good, amazing,...

List of negative words

bad, terrible,
disgusting, sucks,...



Simple threshold classifier

Count positive & negative words in sentence

2

Sushi was great, the food was awesome, but the service was terrible.

If *number of positive words* > *number of negative words*:

$\hat{y} =$

+

Else:

$\hat{y} =$

-

1

Problems with threshold classifier

- How do we get list of positive/negative words?
- Words have different degrees of sentiment:
 - Great > good
 - How do we weigh different words?
- Single words are not enough:
 - *Good* → Positive
 - *Not good* → Negative



Addressed
by learning
a classifier

Addressed
by more
elaborate
features

A (linear) classifier

- Will use training data to learn a weight for each word

Word	Weight
good	1.0
great	1.5
awesome	2.7
bad	-1.0
terrible	-2.1
awful	-3.3
restaurant, the, we, where, ...	0.0
...	...

Scoring a sentence

Word	Weight
good	1.0
great	<u>1.2</u>
awesome	<u>1.7</u>
bad	-1.0
terrible	<u>-2.1</u>
awful	-3.3
restaurant, the, we, where, ...	0.0
...	...

Input x:

Sushi was great,
the food was awesome,
but the service was terrible.

$$\begin{aligned} \text{Score}(x) &= 1.2 + 1.7 - 2.1 \\ &= 0.8 \end{aligned}$$

$$\text{Score}(x) > 0 \Rightarrow +$$

if

$$\text{Score}(x) < 0 \Rightarrow -$$

Called a linear classifier, because output is weighted sum of input.

Word	Weight
...	...



Simple linear classifier

$Score(x)$ = weighted count of words in sentence

If $Score(x) > 0$:

$\hat{y} =$ 

Else:

$\hat{y} =$ 

Sentence
from
review



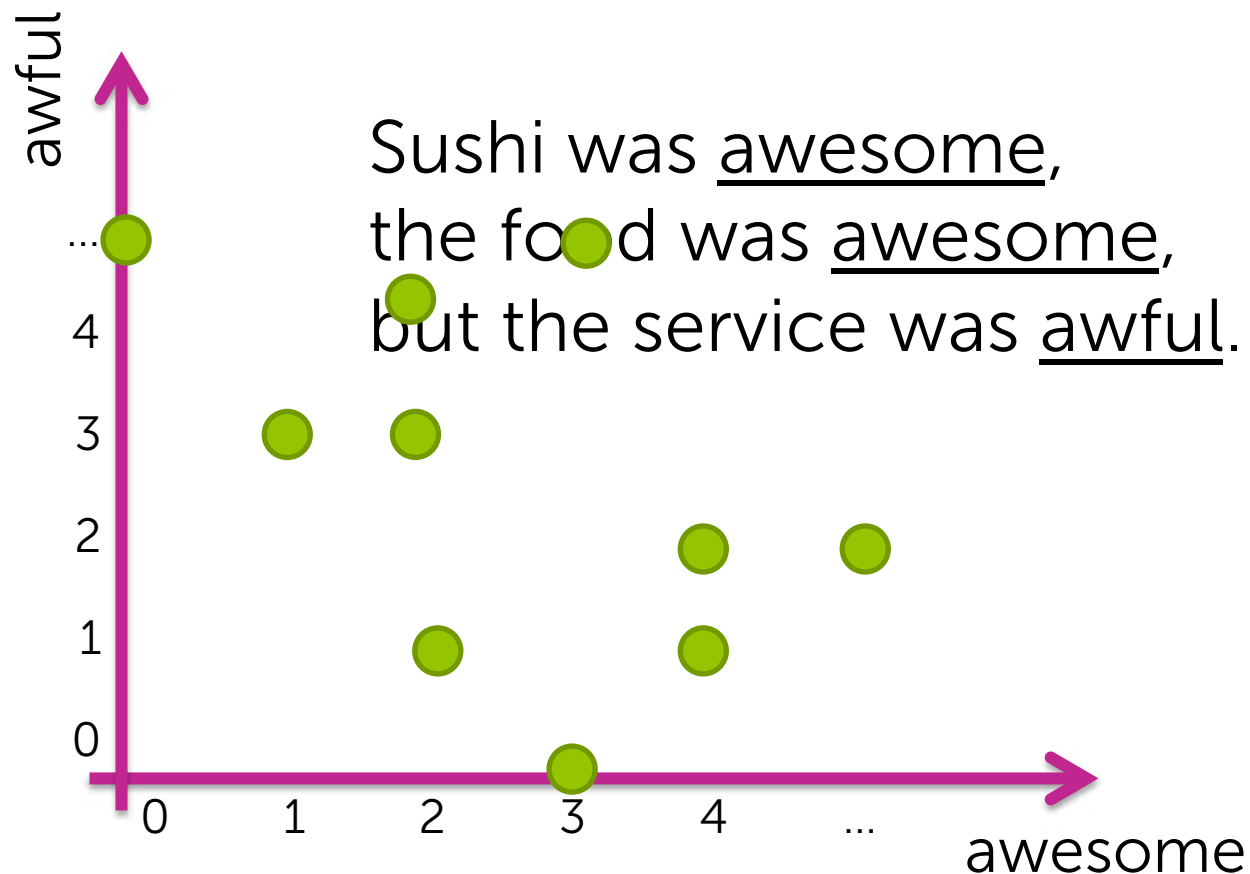
Input: x

Decision boundaries

Suppose only two words had non-zero weight

Word	Weight
awesome	1.0
awful	-1.5

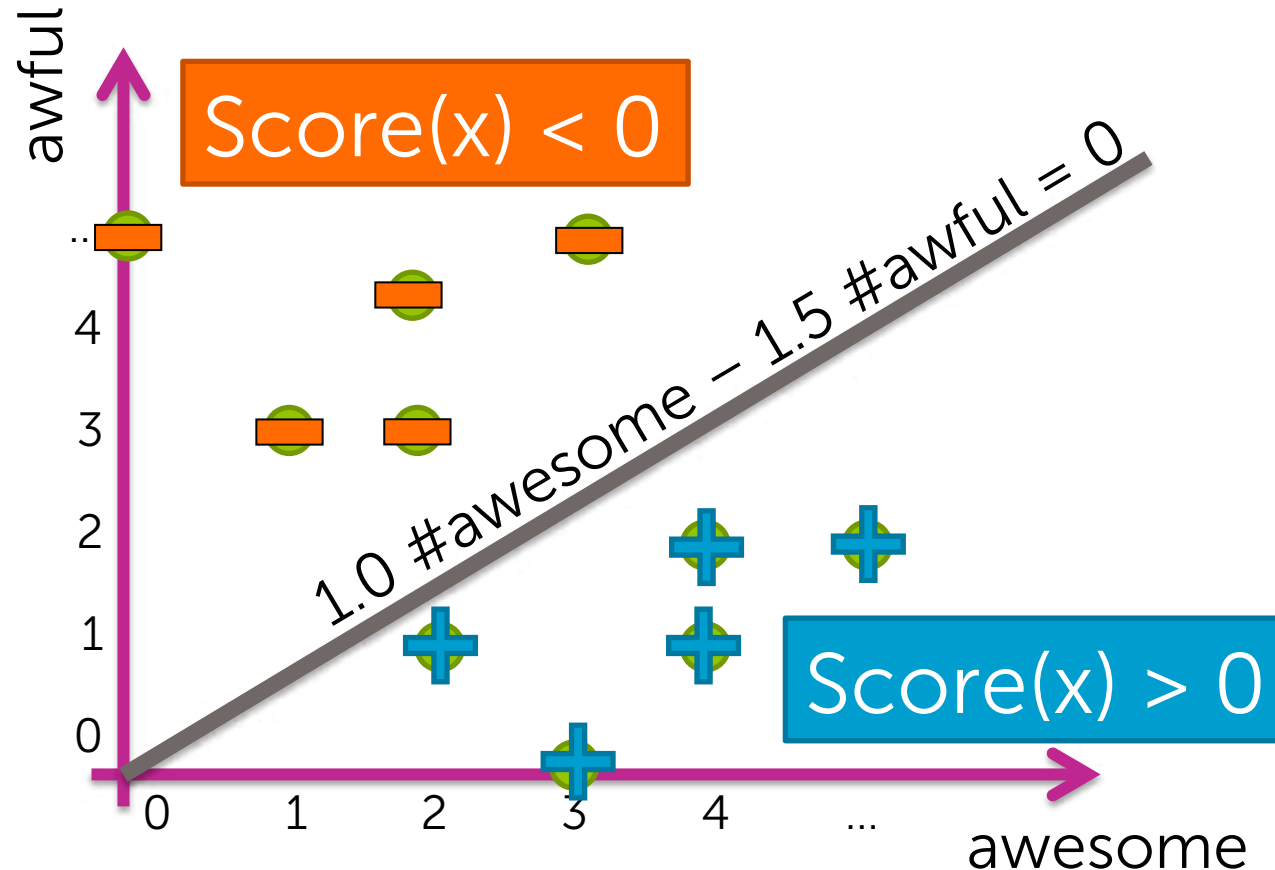
➔ $\text{Score}(x) = 1.0 \# \text{awesome} - 1.5 \# \text{awful}$



Decision boundary example

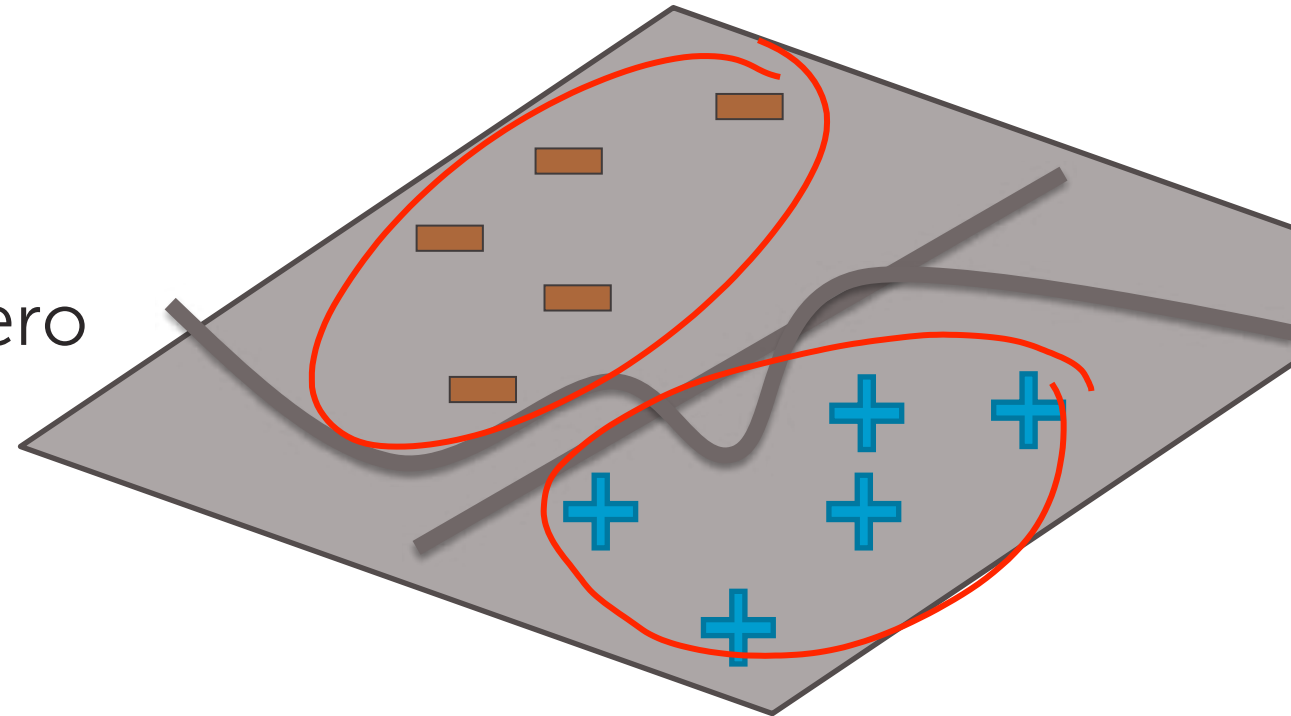
Word	Weight
awesome	1.0
awful	-1.5

➔ $\text{Score}(x) = 1.0 \# \text{awesome} - 1.5 \# \text{awful}$



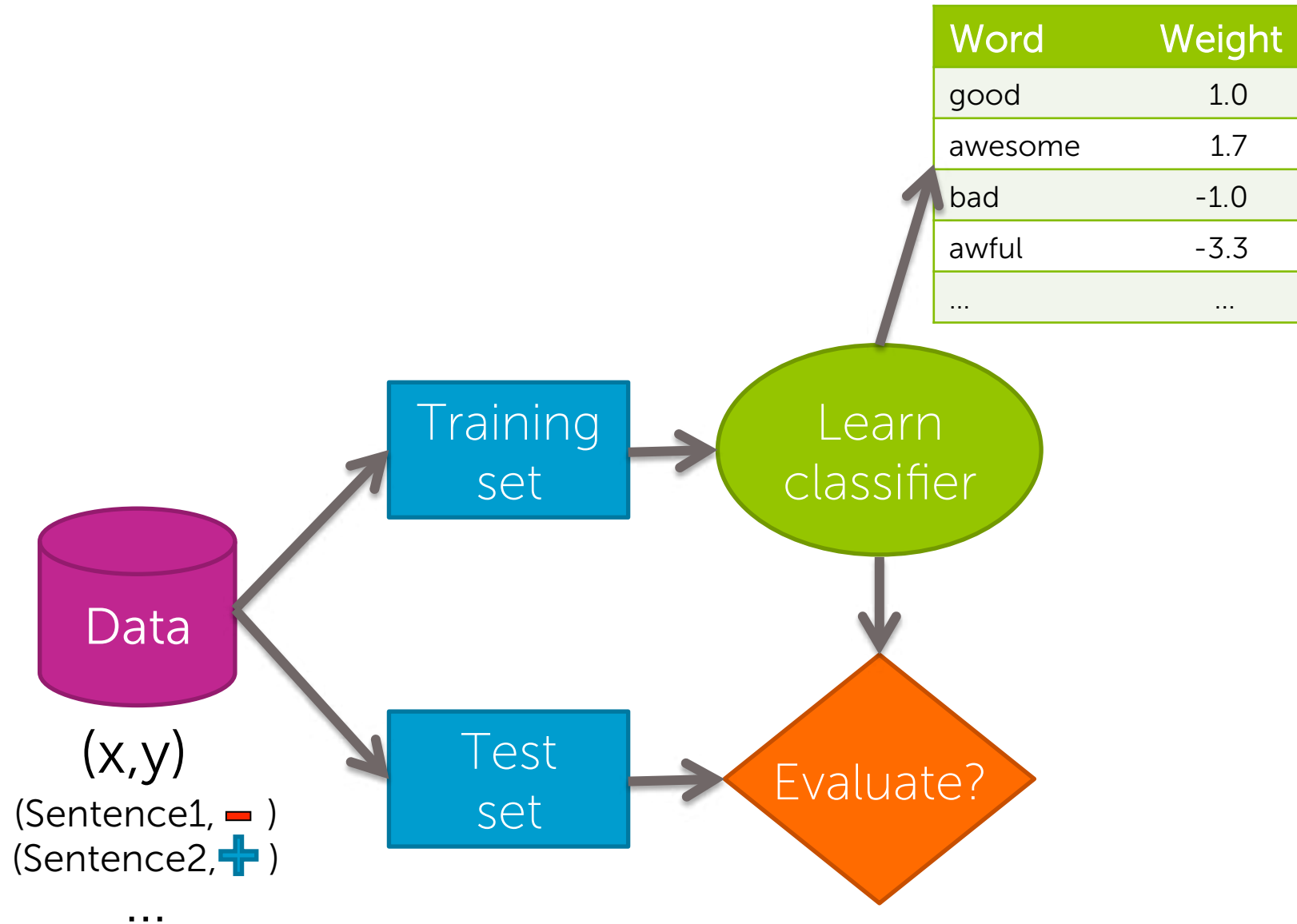
Decision boundary separates positive & negative predictions

- For linear classifiers:
 - When 2 weights are non-zero
→ line
 - When 3 weights are non-zero
→ plane
 - When many weights are non-zero
→ hyperplane
- For more general classifiers
→ more complicated shapes



Training and evaluating a classifier

Training a classifier = Learning the weights



Classification error

Learned classifier

$$\hat{y} = +$$

Test example

(\$Esbid was great, +)

~~Correct!~~
Mistake!

Correct

0

Mistakes

1

Hide label

Classification error & accuracy

- Error measures fraction of mistakes

$$\text{error} = \frac{\# \text{ of mistakes}}{\text{Total \# of sentences}}$$

- Best possible value is 0.0

- Often, measure **accuracy**

- Fraction of correct predictions

$$\text{accuracy} = \frac{\# \text{ of correct}}{\text{Total \# of sentences}}$$

- Best possible value is 1.0

What's a good accuracy?

What if you ignore the sentence, and just guess?

- For binary classification:
 - Half the time, you'll get it right! (on average)
→ accuracy = 0.5
- For k classes, accuracy = $1/k$
 - 0.333 for 3 classes, 0.25 for 4 classes,...

At the very, very, very least,
you should healthily beat random...
Otherwise, it's (usually) pointless...

Is a classifier with 90% accuracy good? Depends...

2010 data shows:
"90% emails sent are spam!"



Predicting every email is spam
gets you 90% accuracy!!!



Majority class prediction



Amazing performance when
there is class imbalance

(but silly approach)

- One class is more common than others
- Beats random (if you know the majority class)

So, always be digging in and asking the hard questions about reported accuracies

- Is there class imbalance?
- How does it compare to a simple, baseline approach?
 - Random guessing
 - Majority class
 - ...
- Most importantly:
what accuracy does my application need?
 - What is good enough for my user's experience?
 - What is the impact of the mistakes we make?

False positives, false negatives, and confusion matrices

Types of mistakes

		<u>Predicted label</u>	
			
<u>True label</u>		True Positive	False Negative (FN)
		False Positive (FP)	True Negative

Cost of different types of mistakes can be different (& high) in some applications

	Spam filtering	Medical diagnosis
False negative	Annoying	Disease not treated
False positive	Email lost <i>Higher cost</i>	Wasteful treatment



Confusion matrix – binary classification

100 test examples

		Predicted label	
			
True label		50	10
		5	35

60

40

$$\text{accuracy} = \frac{85}{100} = 0.85$$

Confusion matrix – multiclass classification

100 test examples

Predicted label

True label	Predicted label		
	Healthy	Cold	Flu
Healthy 70	60	8	2
Cold 20	4	12	4
Flu 10	0	2	8

$$\text{accuracy} = \frac{80}{100} = 0.8$$

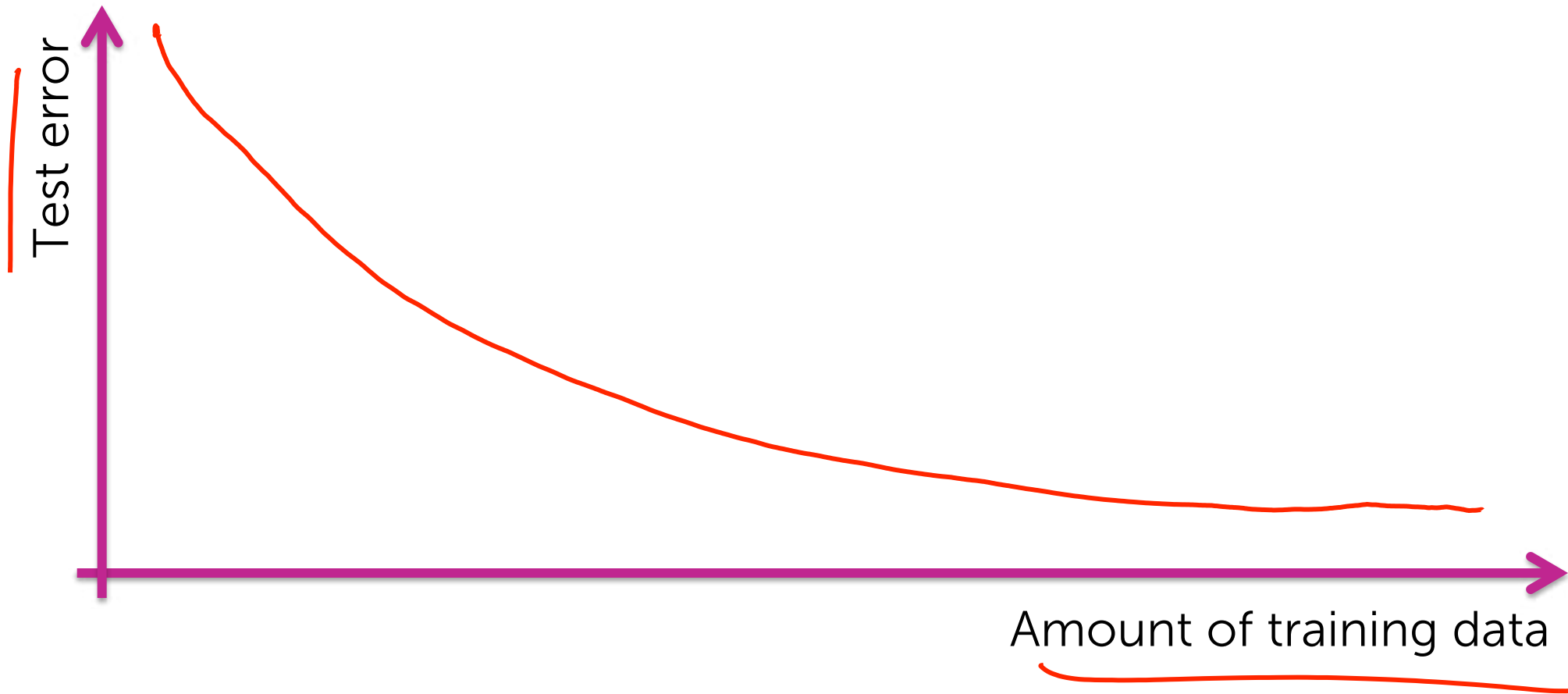
Learning curves:

How much data do I need?

How much data does a model need to learn?

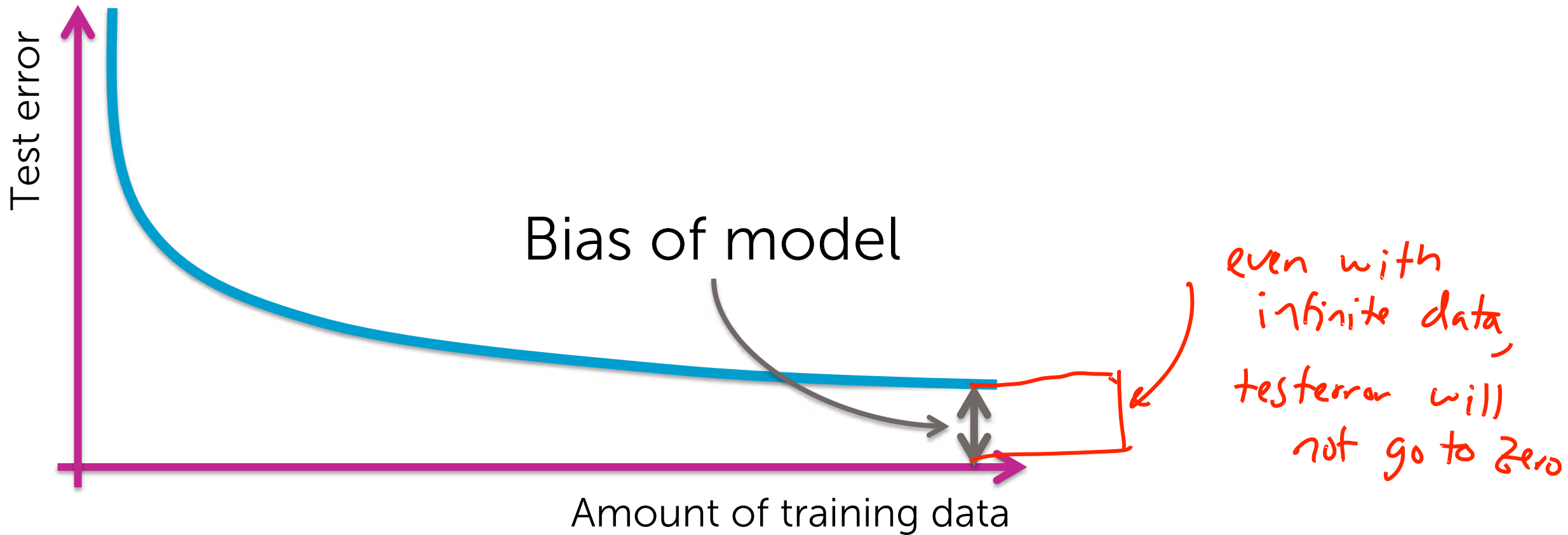
- The more the merrier 😊
 - But data quality is most important factor
- Theoretical techniques sometimes can bound how much data is needed
 - Typically too loose for practical application
 - But provide guidance
- In practice:
 - More complex models require more data
 - Empirical analysis can provide guidance

Learning curves



Is there a limit?

Yes, for most models...



More complex models tend to have less bias...

Sentiment classifier using
single words can do OK, but...



Never classifies correctly:
"The sushi was not good."



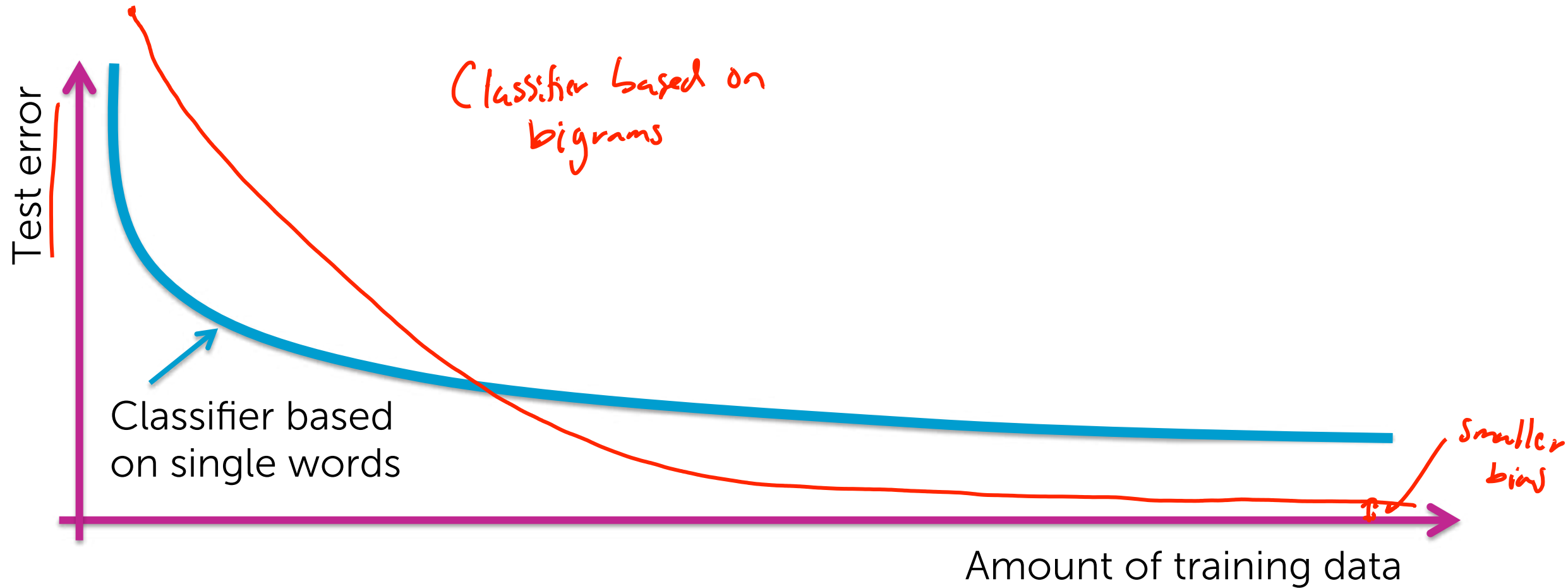
More complex model:
consider pairs of words (bigrams)



Word	Weight
good	+1.5
not good	-2.1

Less bias →
potentially more accurate,
needs more data to learn

Models with less bias tend to need more data to learn well, but do better with sufficient data



Class probabilities

How confident is your prediction?

- Thus far, we've outputted a prediction 
- But, how sure are you about the prediction?
 - *"The sushi & everything else were awesome!"*  $P(y=+|x) = 0.99$
 - *"The sushi was good, the service was OK."*  $P(y=+|x) = 0.55$

Many classifiers provide a confidence level:

$$P(y|x)$$

Output label

Input sentence

Extremely useful in practice



Summary of classification

What you can do now...

- Identify a classification problem and some common applications
- Describe decision boundaries and linear classifiers
- Train a classifier
- Measure its error
 - Some rules of thumb for good accuracy
- Interpret the types of error associated with classification
- Describe the tradeoffs between model bias and data set size
- Use class probability to express degree of confidence in prediction